3 Redistribution Between RIP and OSPF

Step 1: Configure Loopbacks and Interface Addresses

Router R1

```
enable
configure terminal

interface Loopback0
ip address 172.16.1.1 255.255.255.0
exit

interface Loopback48
ip address 192.168.48.1 255.255.255.0
exit

interface Loopback49
ip address 192.168.49.1 255.255.255.0
exit

interface Loopback50
ip address 192.168.50.1 255.255.255.0
exit

interface Loopback51
ip address 192.168.51.1 255.255.255.0
exit

interface Loopback70
ip address 192.168.70.1 255.255.255.0
exit

interface Serial0/0/0
ip address 172.16.12.1 255.255.255.0
clock rate 64000
bandwidth 64
no shutdown
exit

end
```

Router R2

```
enable
configure terminal

interface Loopback0
ip address 172.16.2.1 255.255.255.0
exit

interface Serial0/0/0
ip address 172.16.12.2 255.255.255.0
bandwidth 64
no shutdown
exit

interface Serial0/0/1
ip address 172.16.23.2 255.255.255.0
clock rate 64000
bandwidth 64
no shutdown
exit

end
```

Router R3

```
enable
configure terminal

interface Loopback0
ip address 172.16.3.1 255.255.255.0
exit

interface Loopback20
ip address 192.168.20.1 255.255.255.0
exit

interface Loopback25
ip address 192.168.25.1 255.255.255.0
exit

interface Loopback30
ip address 192.168.30.1 255.255.255.0
exit

interface Loopback35
ip address 192.168.35.1 255.255.255.0
exit

interface Loopback40
ip address 192.168.40.1 255.255.255.0
exit

interface Serial0/0/1
ip address 172.16.23.3 255.255.255.0
bandwidth 64
no shutdown
exit

end
```

Step 2: Configure RIPv2

R1

```
configure terminal
router rip
version 2
no auto-summary
network 172.16.0.0
network 192.168.48.0
network 192.168.49.0
network 192.168.50.0
network 192.168.51.0
network 192.168.70.0
end
```

R2

```
configure terminal
router rip
version 2
no auto-summary
network 172.16.0.0
end
```

Verification

```
show ip route rip
```

Step 3: Configure Passive Interfaces in RIP

R2

```
configure terminal
router rip
passive-interface serial0/0/1
end
show ip protocols
```

R1

```
configure terminal
router rip
passive-interface loopback0
passive-interface loopback48
passive-interface loopback49
passive-interface loopback50
passive-interface loopback51
passive-interface loopback70
end
```

R2 Loopback Passive

```
configure terminal
router rip
passive-interface loopback0
end
```

Step 4: Summarize a Supernet with RIP

```
show ip route rip
configure terminal
ip route 192.168.48.0 255.255.252.0 null0
router rip
redistribute static
end
```

Verification

```
show ip route
```

Step 5: Suppress Routes Using Prefix Lists

```
configure terminal

ip prefix-list RIP-OUT permit 192.168.48.0/22
ip prefix-list RIP-OUT deny 192.168.48.0/22 le 24
ip prefix-list RIP-OUT permit 0.0.0.0/0 le 32

router rip
distribute-list prefix RIP-OUT out serial0/0/0

end
```

Verification

```
show ip route rip
show ip rip database
```

Step 6: Configure OSPF

R2

```
configure terminal
router ospf 1
network 172.16.23.0 0.0.0.255 area 0
end
```

R3

```
configure terminal
router ospf 1
network 172.16.0.0 0.0.255.255 area 0
network 192.168.0.0 0.0.255.255 area 0
end
```

Configure Loopback OSPF Type (R3)

```
configure terminal

interface Loopback0
ip ospf network point-to-point
exit

interface Loopback20
ip ospf network point-to-point
exit

interface Loopback25
ip ospf network point-to-point
exit

interface Loopback30
ip ospf network point-to-point
exit

interface Loopback35
ip ospf network point-to-point
exit

interface Loopback40
ip ospf network point-to-point
exit

end
```

Verification

```
show ip ospf neighbor
show ip route ospf
```

Step 7: Passive Interfaces in OSPF

```
configure terminal
router ospf 1
passive-interface default
no passive-interface serial0/0/1
end
```

Verification

```
show ip protocols
```

Step 8: One-Way Redistribution

```
configure terminal
router rip
redistribute ospf 1 metric 4
end
```

Verification

```
show ip protocols
show ip route rip
```

Test

```
ping 192.168.30.1
traceroute 192.168.30.1
```

Default Route from OSPF

```
configure terminal
router ospf 1
default-information originate always
end
```

Step 9: Redistribute Between RIP and OSPF

Remove default route

```
configure terminal
router ospf 1
no default-information originate always
end
```

Redistribute RIP

```
configure terminal
router ospf 1
redistribute rip
redistribute rip subnets
end
```

Verification

```
show ip route ospf
```